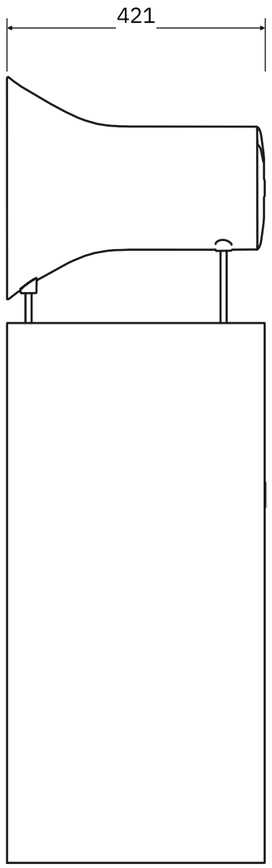
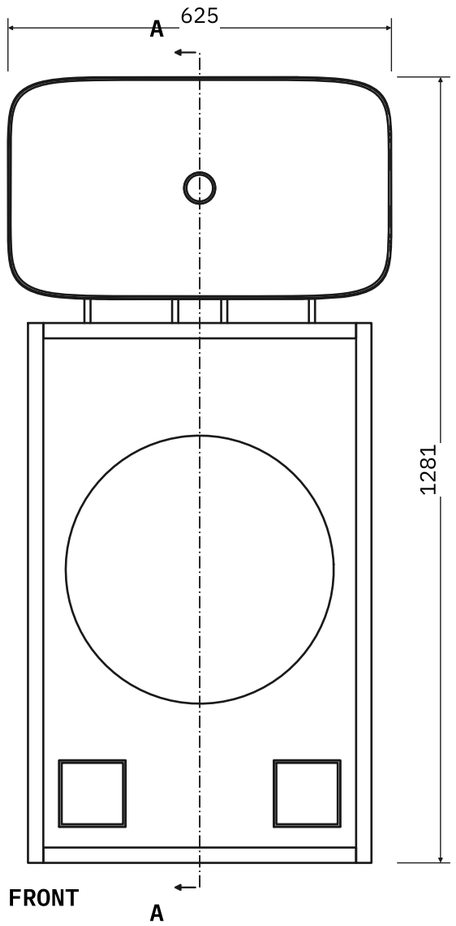
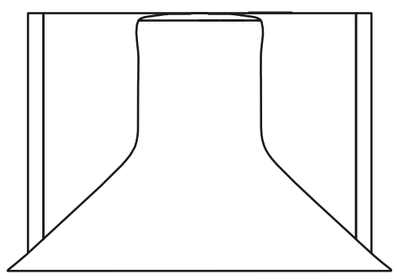
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

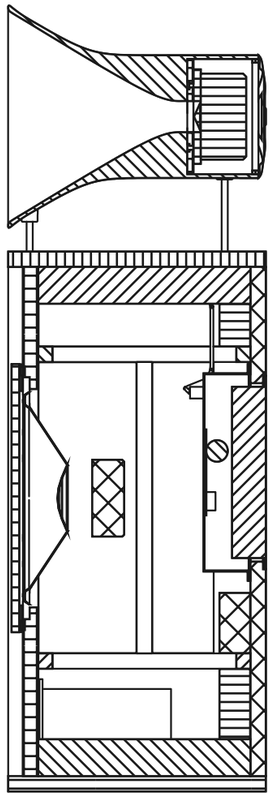
General arrangement

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Section A-A (sheet 1): the product cut at x 280.0 and seen from the right; cut faces hatched, a different angle for each part.



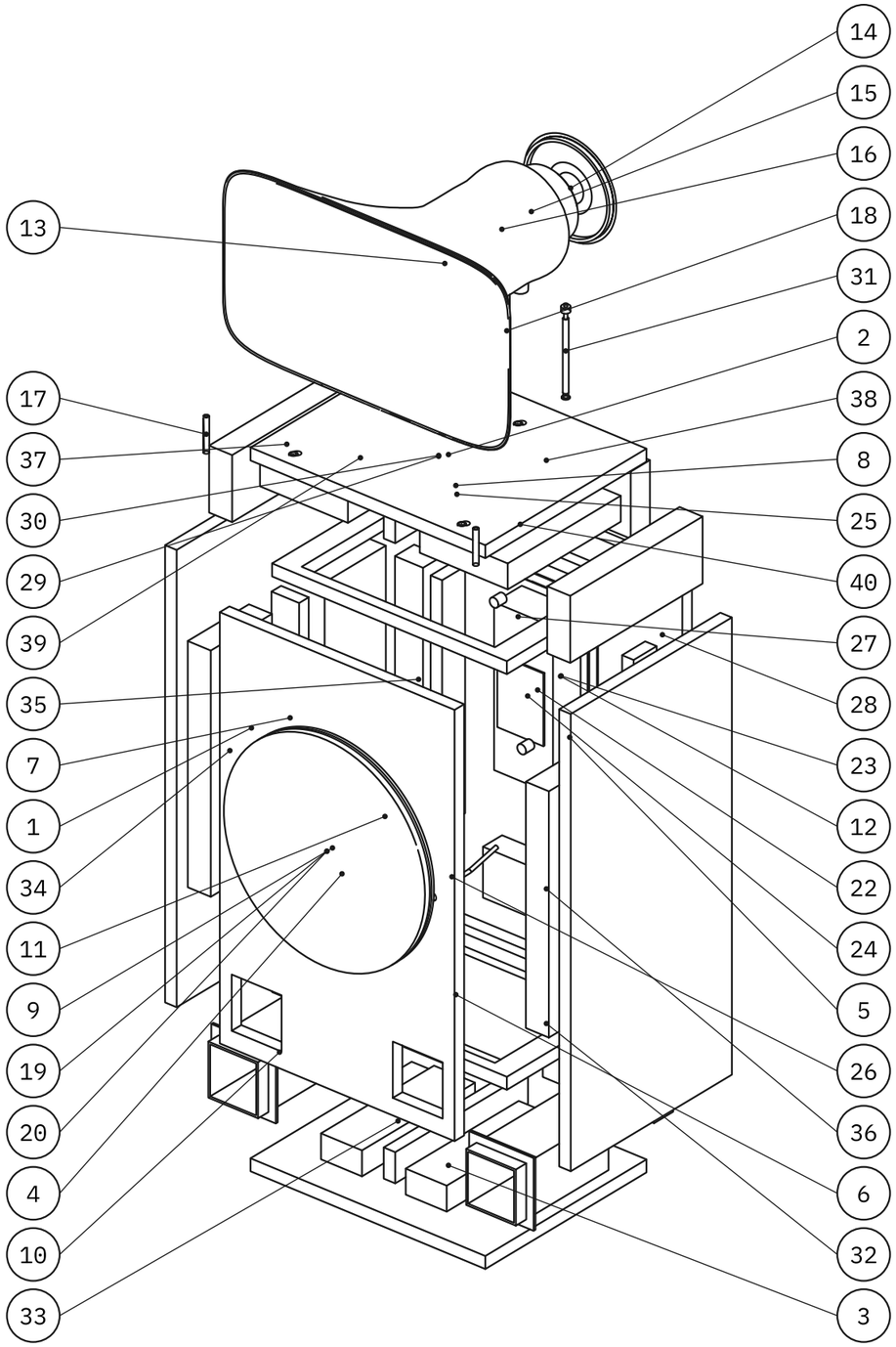
Section A-A

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



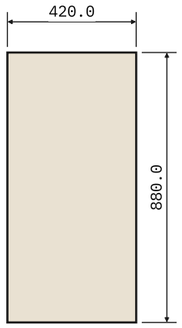
Exploded view

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

mm | issue 2026-10-14 | first angle

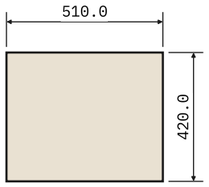
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:20. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



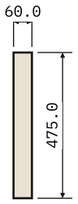
left, right

25 mm recycled-hdpe, 4 for the set



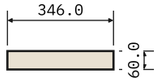
bottom

25 mm recycled-hdpe, 2 for the set



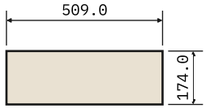
rib-left-1, rib-right-1, rib-back-1-1, rib-back-2-1

25 mm recycled-hdpe, 8 for the set



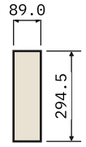
rib-top, rib-bottom

25 mm recycled-hdpe, 4 for the set



lining-back-1

50 mm polyester-batting, 2 for the set



lining-bottom-1a, lining-bottom-1b

50 mm polyester-batting, 4 for the set

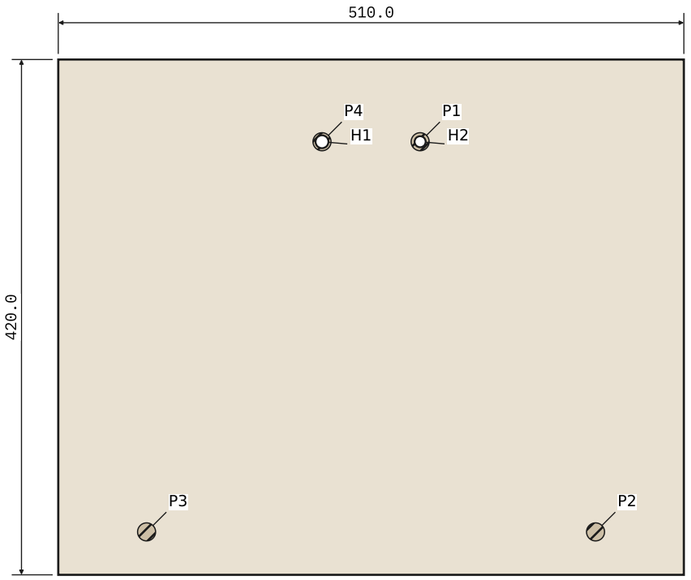
Sheet parts: plain panels

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/top.dxf. Marks listed at the right.



top: 25 mm recycled-hdpe, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø14.7	295.0	353.0	15 from face
P2	Ø14.7	438.0	35.0	15 from face
P3	Ø14.7	72.0	35.0	15 from face
P4	Ø14.7	215.0	353.0	15 from face
H1	Ø10.5	215.0	353.0	through
H2	Ø9.0	295.0	353.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

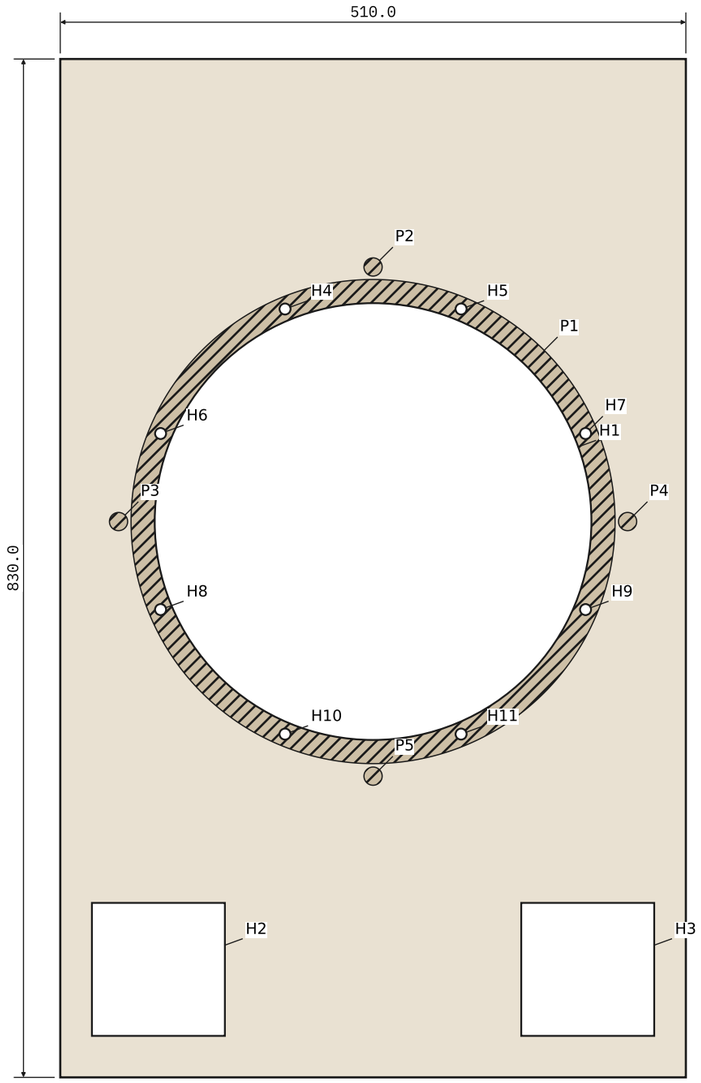
Sheet part: top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/front-panel.dxf. Marks listed at the right.



front-panel: 25 mm recycled-hdpe, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø394.6	255.0	453.0	11 from face
P2	Ø14.9	255.0	660.5	3.2 from face
P3	Ø14.8	47.5	453.0	3.2 from face
P4	Ø14.8	462.5	453.0	3.2 from face
P5	Ø14.8	255.0	245.5	3.2 from face
H1	Ø356.0	255.0	453.0	through
H2	108.4 x 108.4	80.0	88.0	through
H3	108.4 x 108.4	430.0	88.0	through
H4	Ø9.0	183.2	626.2	through
H5	Ø9.0	326.8	626.2	through
H6	Ø9.0	81.8	524.8	through
H7	Ø9.0	428.2	524.8	through
H8	Ø9.0	81.8	381.2	through
H9	Ø9.0	428.2	381.2	through
H10	Ø9.0	183.2	279.8	through
H11	Ø9.0	326.8	279.8	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

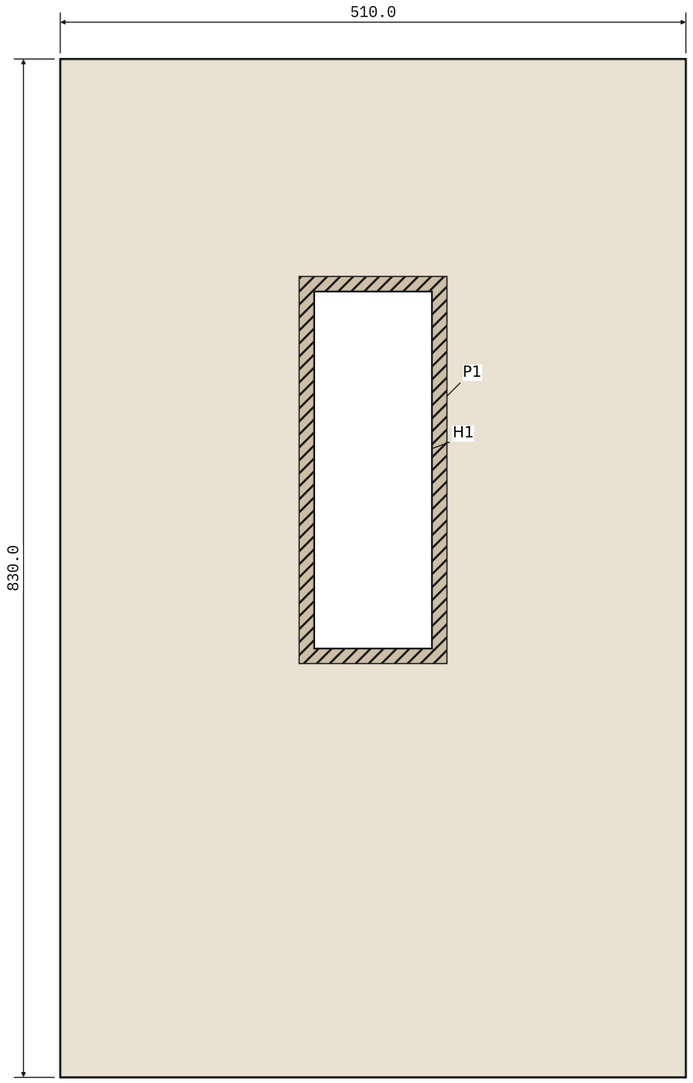
Sheet part: front-panel

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/back.dxf. Marks listed at the right.



back: 25 mm recycled-hdpe, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	120.5 x 315.5	255.0	495.0	4.5 from face
H1	96.0 x 291.0	255.0	495.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: back

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

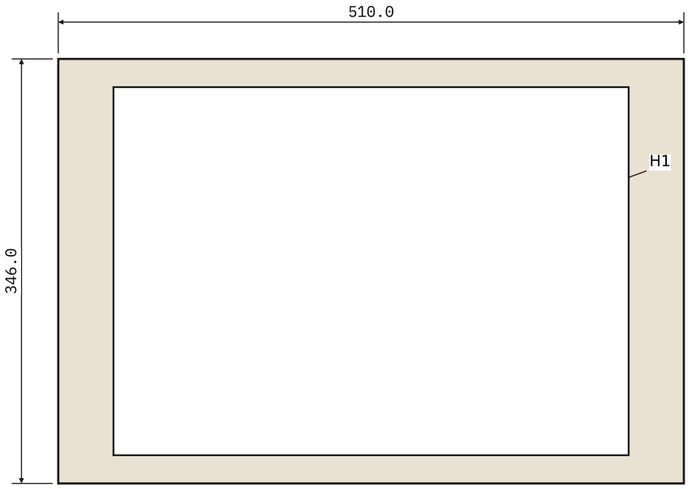
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	420.0 x 300.0	255.0	173.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



brace: 25 mm recycled-hdpe, 2 for the set, seen from its outer face

Sheet part: brace

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

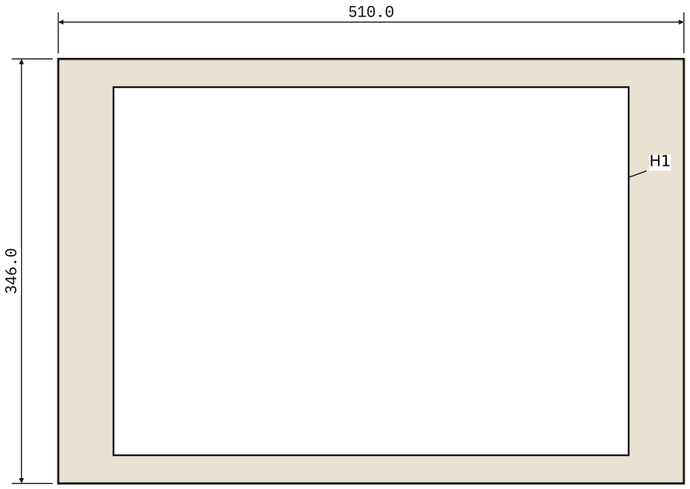
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace-2.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	420.0 x 300.0	255.0	173.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



brace-2: 25 mm recycled-hdpe, 2 for the set, seen from its outer face

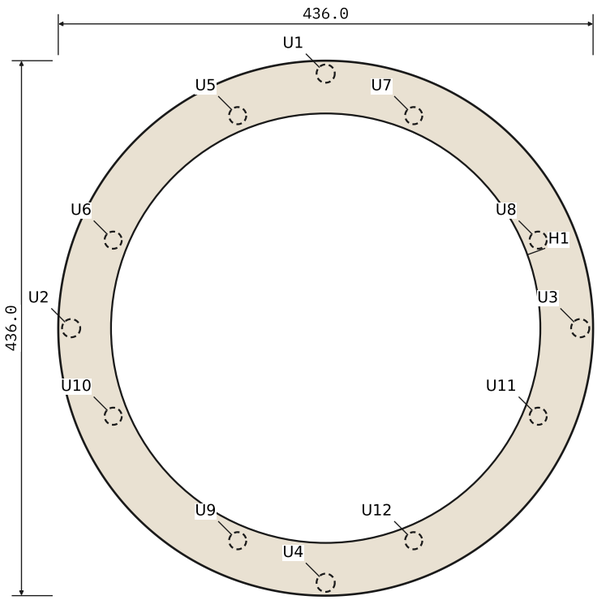
Sheet part: brace-2

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/woofer-ring.dxf. Marks listed at the right.



woofer-ring: 6 mm recycled-hdpe, 2 for the set, seen from its outer face

inner-face pockets: dxf/woofer-ring-underside.dxf (seen from that face)

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø14.9	218.0	425.5	3.2 from inside
U2	Ø14.9	10.5	218.0	3.2 from inside
U3	Ø14.8	425.5	218.0	3.2 from inside
U4	Ø14.8	218.0	10.5	3.2 from inside
U5	Ø14.0	146.3	391.2	4 from inside
U6	Ø14.0	44.8	289.8	4 from inside
U7	Ø14.0	289.8	391.2	4 from inside
U8	Ø14.0	391.2	289.8	4 from inside
U9	Ø14.0	146.3	44.8	4 from inside
U10	Ø14.0	44.8	146.2	4 from inside
U11	Ø14.0	391.2	146.2	4 from inside
U12	Ø14.0	289.8	44.8	4 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: woofer-ring

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

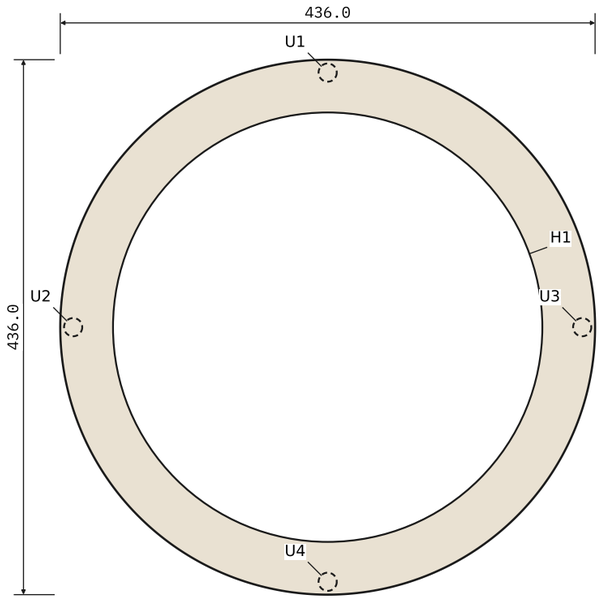
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/disc-frame.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø14.9	218.0	425.5	3.2 from inside
U2	Ø14.9	10.5	218.0	3.2 from inside
U3	Ø14.8	425.5	218.0	3.2 from inside
U4	Ø14.8	218.0	10.5	3.2 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-frame: 12 mm recycled-hdpe, 2 for the set, seen from its outer face

inner-face pockets: dxf/disc-frame-underside.dxf (seen from that face)

Sheet part: disc-frame

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

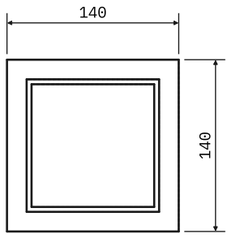
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

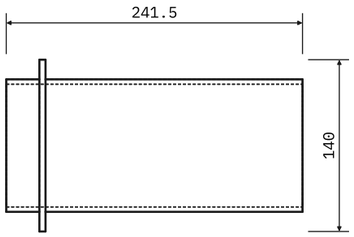
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

10. port-left, port-right: asa print, on its flange; 442 cm3 each; 4 for the set; 1:5

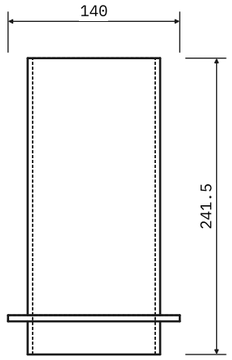
100 square, 242 long from the panel's face: the 2 together tune the box to 38 Hz; print 15 longer and trim to tune; fitted from inside, its 140 square flange on a 2 closed-cell PE foam seal on the panel's back, four thread-forming screws for plastics through it



FRONT



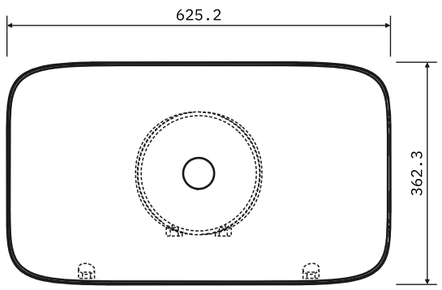
FROM THE RIGHT



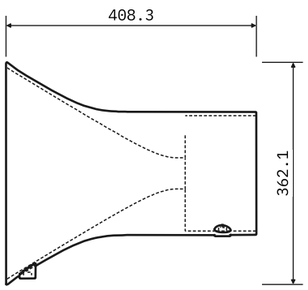
FROM ABOVE

13. waveguide: cast, cast-resin; 5222 cm3 each; 2 for the set; 1:10

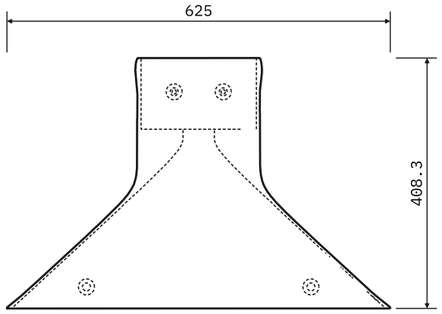
mouth 610 x 346, 278 deep, 2 in throat; a 10 rim rounded right round, 625 x 362 outside; a shell 6 thick following its flare; the driver in a pod 200 across behind, closed by a cap; four pads under it take the posts' brass inserts; cast in tinted clear resin in a two-part silicone mould from a CNC-cut plug, its faces polished satin (a slow deep-pour resin: the neck round the throat is solid, 5.2 L in all), or pigmented the same way for the opaque and marbled finishes



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

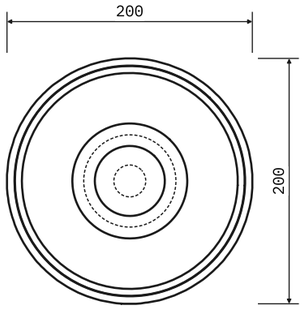
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

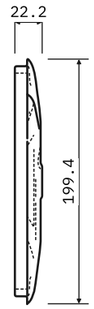
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

14. waveguide-cap: cast, cast-resin; 187 cm3 each; 2 for the set; 1:5

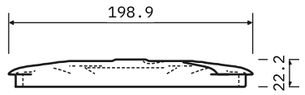
closes the pod behind the driver; a spigot locates it, three M4 screws hold it; the HF pair leaves through a post below; unplugged here, the horn lifts off its sockets



FRONT



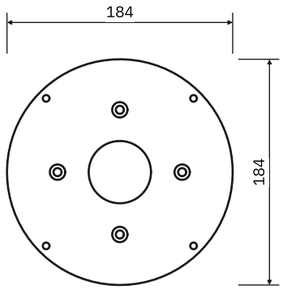
FROM THE RIGHT



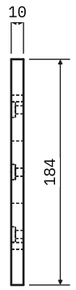
FROM ABOVE

16. hf-adaptor: lathe, aluminium; 242 cm3 each; 2 for the set; 1:5

184 across, 10 thick, bored 50.8 (the throat); 4 countersunk M6 on 101.6 take the driver from the front; 4 x 5.5 on 170 (at 45 degrees to them) take M5 socket caps into the brass inserts in the horn's boss; black anodised



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

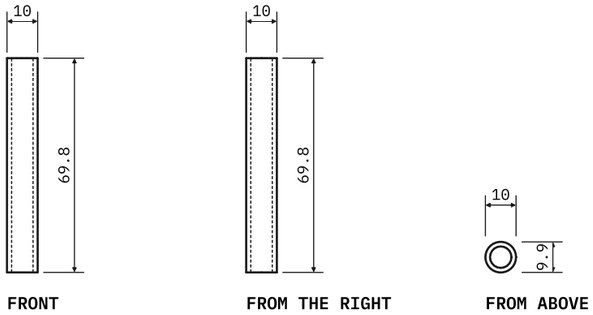
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

17. post-1, post-2: lathe, stainless; 3 cm3 each; 4 for the set; 1:2

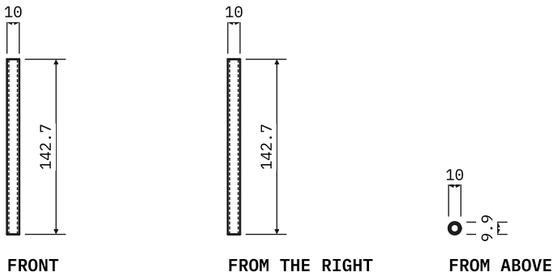
70 long (49 showing): stainless tube 10 x 7, threaded M10 x 1 8 at the top into the brass insert in its pad; its foot plain, standing 13 deep in a polyurethane sleeve in a brass socket in the top; its bore deburred



FRONT FROM THE RIGHT FROM ABOVE

18. post-3, post-4: lathe, stainless; 6 cm3 each; 4 for the set; 1:5

143 long (118 showing): stainless tube 10 x 7, threaded M10 x 1 12 at the top into the brass insert in its pad; its foot plain, standing 13 deep in a polyurethane sleeve in a brass socket in the top; its bore deburred; the HF lead runs down inside it



FRONT FROM THE RIGHT FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in pressed recycled plastic, no wood

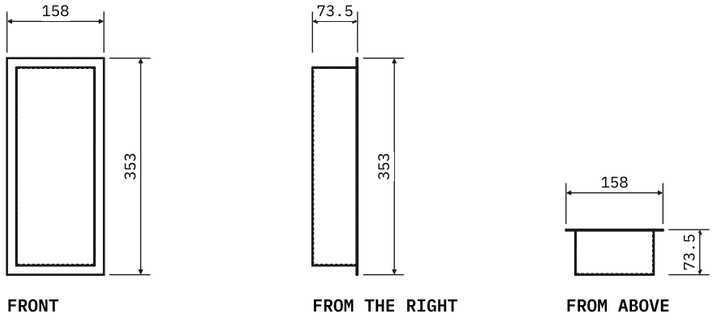
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

22. amp-box: sheet metal, aluminium; 180 cm3 each; 2 for the set; 1:10

125 x 320 inside, 75 deep from the back: 1.5 aluminium folded, its corners welded, black anodised; a 15 flange round its open side, on a 3 EPDM gasket on the back's inner face, ten screws through it, so the box stays airtight and the mains electronics are enclosed in metal; two M12 cable glands in its front; re-trim the ports for the volume it takes



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For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	left, right	cut	4	25 mm Recycled HDPE sheet, pressed to order (a sheet press workspace)	420 x 880	[PRICE]
2	top	cut	2	25 mm Recycled HDPE sheet, pressed to order (a sheet press workspace)	510 x 420	[PRICE]
3	bottom	cut	2	25 mm Recycled HDPE sheet, pressed to order (a sheet press workspace)	510 x 420	[PRICE]
4	front-panel	cut	2	25 mm Recycled HDPE sheet, pressed to order (a sheet press workspace)	510 x 830	[PRICE]
5	back	cut	2	25 mm Recycled HDPE sheet, pressed to order (a sheet press workspace)	510 x 830	[PRICE]
6	brace, brace-2	cut	4	25 mm Recycled HDPE sheet, pressed to order (a sheet press workspace)	510 x 346	[PRICE]
7	rib-left-1, rib-right-1, rib-back-1-1, rib-back-2-1	cut	8	25 mm Recycled HDPE sheet, pressed to order (a sheet press workspace)	60 x 475	[PRICE]
8	rib-top, rib-bottom	cut	4	25 mm Recycled HDPE sheet, pressed to order (a sheet press workspace)	346 x 60	[PRICE]
9	woofer-ring	cut	2	6 mm Recycled HDPE sheet, pressed to order (a sheet press workspace)	436 x 436	[PRICE]
10	port-left, port-right	print	4	ASA, FDM (outdoor, paintable)	442 cm3	38.91
11	woofer	buy	2	FaitalPRO 15PR400-8 15 in woofer (treated paper cone, neodymium, 8 ohm)	393 frame, 356 cut-out, 165 deep	194.00
12	amp-plate	buy	2	Hypex FusionAmp FA122 DSP plate amplifier, 2 x 125 W into 4 ohm	315 x 120 x 55 plate, 291 x 96 recess	422.40
13	waveguide	cast	2	Casting resin (polyurethane or epoxy), tinted clear, translucent or pigmented	5222 cm3, 635 x 418 x 372	684.78
14	waveguide-cap	cast	2	Casting resin (polyurethane or epoxy), tinted clear, translucent or pigmented	187 cm3, 210 x 32 x 210	138.21
15	hf	buy	2	B&C Speakers DE1085TN 2 in compression driver, titanium diaphragm, 100 mm voice coil, neodymium, 16 ohm (round 5: the owner's bigger throat)	154 across, 86 deep, 4 x M6 on 102	512.00
16	hf-adaptor	lathe	2	Aluminium 6061	194 x 20 x 194	43.82
17	post-1, post-2	lathe	4	Stainless steel 304	20 x 20 x 80	15.96
18	post-3, post-4	lathe	4	Stainless steel 304	20 x 20 x 153	16.83
19	disc-frame	cut	2	12 mm Recycled HDPE sheet, pressed to order (a sheet press workspace)	436 x 436	[PRICE]
20	cloth-disc	buy	2	Acoustically transparent speaker cloth (a knit sold for grilles), about 0.6 x 0.95 m a grille, in the finish's colour		[PRICE]
21	cloth-disc-magnets	buy	2	The ring's and the disc's magnets: eight 15 x 3 neodymium discs (N42) and four 15 x 3 mild-steel discs, a set: four magnets in the panel and the steel discs in the ring's back hold the ring, four in the disc's frame hold the disc to the same		[PRICE]
22	amp-box	sheet metal	2	Aluminium 6061	305 x 500 x 2	85.72

31 more lines: bom.md

Notes

1. The waveguide stands on four posts, stainless lamp tube 10 x 7, each threaded M10 x 1 at its top into a brass insert bonded in a pad cast under the horn (drilled after casting, through a jig taken off the top's sockets), its plain foot standing in a polyurethane sleeve in a brass socket pressed into the top, so the horn and its posts lift off as one. An M6 rod down the free rear post, from a thumb nut inside the pod to an insert under the top, holds them down. The HF lead, a round silicone cable 5 across, runs from the amplifier's compartment up the back, through a gland under the top and up inside a rear post into the pod, where it plugs into the driver's leads. To lift the horn off: the cap off, the nut undone and the lead unplugged.
2. The FA122 crosses at about 800 Hz (LR4: B&C asks 12 dB/oct or steeper there) with the HF channel about 9 dB down (108 dB at 2.83 V against the woofer's 99), part of it in the passive L-pad after the amplifier, so its hiss stays inaudible: two 25 W aluminium-housed resistors on a plate in the compartment, behind a 33 uF film capacitor that keeps low frequencies off the driver if the preset is ever lost.

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